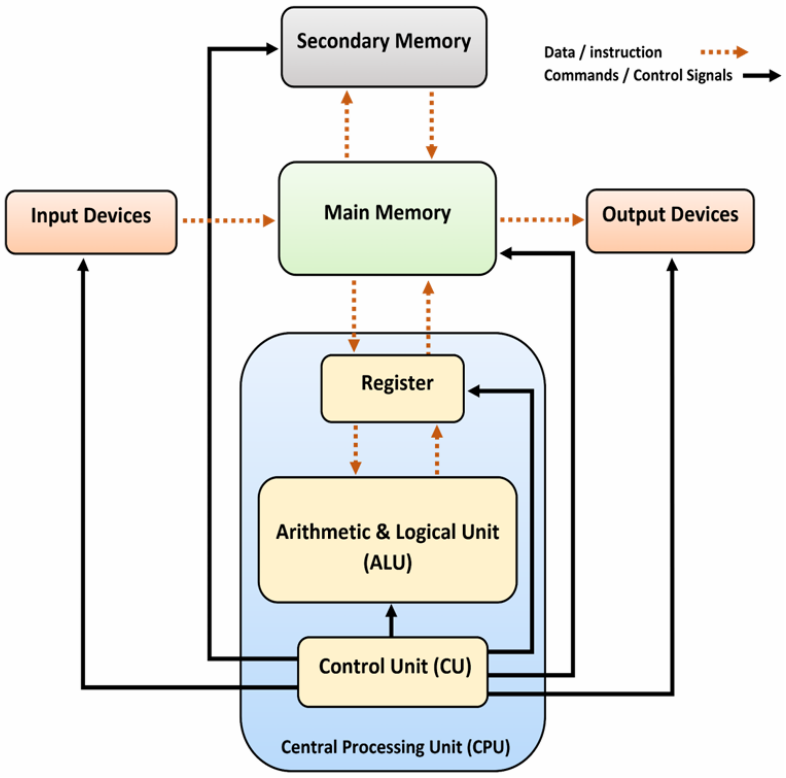
CHAP1 : INTRODUCTION TO COMPUTER SCIENCE

Introduction to Computer System

A computer is an electronic device that takes instructions and data from a user as input, processes them, and produces some meaningful output as a result. Computers are a combination of hardware and software. Computer systems work on the IPO (Input-Process-Output) Model. They process information stored as binary digits (0s and 1s) and perform complex tasks with incredible speed and accuracy.



Computer Block Diagram

Input Devices

- The hardware devices that send input data from the user to the CPU are known as Input Devices.
- They are responsible for capturing, and transferring data from the outside world into the computer system.
- Some of the popular Input Devices are Keyboard, Mouse, Joystick, Scanner, Microphone, Touchpad, Biometric Sensors, Graphic Tablet, Bar/QR Code Readers, Webcam, Magnetic Ink Character Reader (MICR) and Optical Character Reader (OCR) devices etc.



Output Devices

- The hardware devices that are responsible for displaying output from the computer are known as Output Devices.
- Some of the popular Output Devices are Display Monitor, Projector, Speaker, Printer, Plotter, Headphones etc



Central Processing Unit

- The Central Processing Unit (CPU) is the brain of the computer.
- It is responsible for processing the instructions received, and generating the results.
- The CPU contains three major components, namely an Arithmetic and Logic Unit (ALU), a Control Unit (CU) and a set of Registers.
- The ALU performs arithmetic operations such as addition, subtraction, division, multiplication, exponentiation etc. It is also responsible for performing logical operations that result in either True or False.
- The Control Unit (CU) is responsible for proper operation of the system. It controls the different operations of the computer by generating control signals.
- The set of Registers are local/temporary storage for the CPU.
- Registers can be accessed in a single CPU instruction and thus are extremely fast storage.
- Registers are limited in number and thus data has to be transferred from the main/cache memory to the registers for processing of data.

Memory

1. The location where the computer stores data, instructions, and programs is called the Memory.
2. Computer memory is organized into memory cells, each of which stores a fixed amount of data, typically represented in binary as 0s and 1s.
3. The Memory of a computer is divided into Primary, Cache and Secondary Memory.

- Primary Memory 1: The primary memory is directly accessible by the CPU, It is a high speed memory which can closely match the operational speed of the CPU. There are two kinds of primary memory, namely RAM and ROM.

- Secondary Memory: Secondary memory is also known as permanent memory. It is non-volatile in nature, i.e the data does not get lost even after power is turned off from the computer. Examples of Secondary memory are Hard Disk Drive, Solid State Drive, USB Flash Drive.

Common words and expressions related to computer are:

Data: is what we feed into the computer. There are raw facts (text, image, sound, …) that still need to be processed (organized and arranged) into a form that can be understood and used, and later stored. Examples: population statistics, blood samples in a hospital, pictures, …

Data processing: This is the term used when a computer follows the directives of programs installed in it to transform the input data to obtain meaningful output. The processing is handled mainly by the processor of the computer system.

Information: This is meaningful and useful material that is derived from the processing of data. Information is what comes out as output.

Computing: This is the act of developing and using the computer to perform an activity. It includes the designing and building of hardware and software systems for a wide range of purposes, processing and managing various kind of information, doing scientific studies using computers, …

Computer science: It is the study of computers and their applications, including their design, operation, and use in processing information.

ICT: stands for Information and Communication Technologies. It is the application of modern communications, computing and digital technologies to create, store, exchange and use information in it various forms.

Program: Set of instructions that enable a computer to perform a given task.

Computer user: That is somebody with little or no technical knowledge of computers, but who can use the computer to create specific documents and to communicate.

Computer Literacy: It is the knowledge and ability to use computers and technologies efficiently.

Advantages of Computers

- Multi-tasking - can perform multiple tasks simultaneously
- Speed - computes data rapidly
- Productivity - automates and streamlines work process to increase productivity
- Accuracy - tasks are performed with high accuracy
- Connectivity - access to vast amount of data and global communication networks
- Education- enhances learning through stores resources
- Storage - can store large amounts of data for a long time
- Reliability - can perform its functions adequately in a controlled environment with high precision

Disadvantages of Computers

- High initial costs of purchase, maintenance
- Long use may lead to health issues
- Security Risks - vulnerable to hacking, loss and theft of information
- Environmental Impact due to high carbon emissions
- Software issues - problems may arise due to buggy software

I. IMPORTANCE OF COMPUTER SCIENCE

Computer Science is important because it:

- Simplifies Work: Computers can do tasks faster and more accurately than humans.
- Saves Time: Sending an email or message is faster than writing letters.
- Stores Information: Computers can store large amounts of data safely.
- Improves Communication: Through the internet, people can connect easily.
- Helps in Decision Making: Software helps analyze data and make smart decisions.
- Creates Career Opportunities: Many jobs today require computer knowledge.

II. APPLICATIONS OF COMPUTER SCIENCE IN REAL LIFE

Field	Application Example
Education	Online learning, e-libraries, digital exams
Health	Patient records, hospital management systems
Banking	ATMs, online banking, data security
Business	Payroll systems, accounting software, e-commerce
Transportation	GPS, online ticket booking
Communication	Emails, video calls, social media

III. History of Computers

Blaise Pascal invents the first commercial calculator, a hand powered adding machine.

In 1946, ENIAC, based on John Von Neuman model completes.The first commercially successful computer is IBM 701.

A generation refers to the state of improvement in the development of a product. This term is also used in the different advancements of computer technology. With each generation, the circuitry has gotten smaller and more advanced than the previous generations before it. As a result of the miniaturization, the speed, power and memory of computers has proportionally increased. New discoveries are constantly being developed that affect the way we live, work and play. In terms of technological developments over time, computers have been broadly classed into five generations.

1. The First Generation (1940-1956)

The first computers used vacuum tubes for circuitry and magnetic drums for memory, and were often enormous, taking up entire rooms. They were very expensive to operate and in addition to using a great deal of electricity, they generated a lot of heat, which was often the cause of malfunctions. First generation computers relied on machine language to perform operations, and

they could only solve one problem at a time. Input was based on punched cards and paper tape, and output was displayed on printouts.

The computers UNIVAC , ENIAC of the US and BESEM of the former Soviet Union are examples of first-generation computing devices.

2. The Second Generation (1956-1963)

Transistors replaced vacuum tubes and ushered in the second generation of computers. Computers become smaller, faster, cheaper, more energy-efficient and more reliable than their first generation predecessors. Second-generation computers still relied on punched cards for input and printouts for output. High-level programming languages were being developed, such as early versions of COBOL and FORTRAN.

The first computers of this generation were developed for the atomic energy industry.

The computers IBM-1070 of the US and MINSK of the former Soviet Union belonged to the second generation.

3. The Third Generation (1964-1971): Integrated Circuits

The development of the integrated circuit was the hallmark of the third generation of computers.

Transistors were miniaturized and placed on silicon chips, called semiconductors, which drastically increased the speed and efficiency of computers. Users interacted with third generation computers through keyboards and monitors and interfaced with an operating system, which allowed the device to run many different applications at one time. Typical computers of the third generation are IBM 360 (United States) and EC (former Soviet Union).

4. The Fourth Generation (1971-Present): Microprocessors

The microprocessor brought the fourth generation of computers, as thousands of integrated circuits were built onto a single silicon chip. What in the first generation filled an entire room could now fit in the palm of the hand. The Intel 4004 chip, developed in 1971, located all the components of the computer - from the central processing unit and memory to input/output controls - on a single chip.

In 1981 IBM introduced its first computer for the home user, and in 1984 Apple introduced the Macintosh. Microprocessors also moved out of the realm of desktop computers and into many areas of life as more and more everyday products began to use microprocessors.

As these small computers became more powerful, they could be linked together to form networks, which eventually led to the development of the Internet. Fourth generation computers also saw the development of GUI (Graphic User Interface), the mouse and handheld devices.

5. The Fifth Generation (Present and Beyond): Artificial Intelligence

Fifth generation computing devices, based on artificial intelligence, are still in development, though there are some applications, such as voice recognition, that are being used today. The use of parallel processing and superconductors is helping to make artificial intelligence a reality. Quantum computation and molecular and nanotechnology will radically change the face of computers in years to come. The goal of fifth-generation computing is to develop devices that respond to natural language input and are capable of learning and self-organization.

IV. Classification of Computers

Computers are available in different shapes, sizes and weights, due to these different shapes and sizes they perform different sorts of jobs from one another.

1. Mainframe and Super Computers

The biggest in size, the most expensive in price than any other is classified and known as super computer. It can process trillions of instructions in seconds. Governments specially use this type of computer for their different calculations and heavy jobs. This kind of computer is also helpful for forecasting weather reports worldwide.

Another giant in computers after the super computer is Mainframe, which can also process millions of instruction per second and capable of accessing billions of data. This computer is commonly used in big hospitals, airline reservations companies, and many other huge companies prefer mainframe because of its capability of retrieving data on a huge basis. This is normally too expensive and out of reach from a salary-based person who wants a computer for his home.

2. Minicomputers

This computer offers less than mainframe in work and performance. These are the computers, which are mostly preferred by the small type of business personals, colleges, and so on.

3. Microcomputers

These computers are lesser in cost than the computers given above and also, small in size; They can store a big amount of data and have a memory to meet the assignments of students and other necessary tasks of business people. There are many types of microcomputers: desktop, workstation, laptop, PDA, etc.

V. NUMBER SYSTEMS AND DATA REPRESENTATION

Computer must not only be able to carry out computations, they must be able to do them quickly and efficiently. There are several data representations, typically for integers, real numbers, characters, and logical values.

1. Number Systems

A number system is a set of symbols and rules used to represent numbers. The number of different symbols used in a given number system is known as the base or radix of the system. The largest value of a symbol (digit) in a given number system is always less than the base or radix of that system. If the base of a system is represented by "$\boldsymbol{b}$", then the largest value a digit in that system can assume is "$\boldsymbol{b} - \mathbf{1}$".

Examples of number systems include the binary, octal, decimal (denary) and hexadecimal systems.

a. Decimal System

A decimal number uses base-10 for representing numbers. The numbers 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9 are the digits present in the system. Every number can be represented as a sum of powers of 10, with each power weighted by a digit.

For example, in the decimal system, 534 can be represented as

$$534 = 5 \times 10^2 + 3 \times 10^1 + 4 \times 10^0$$

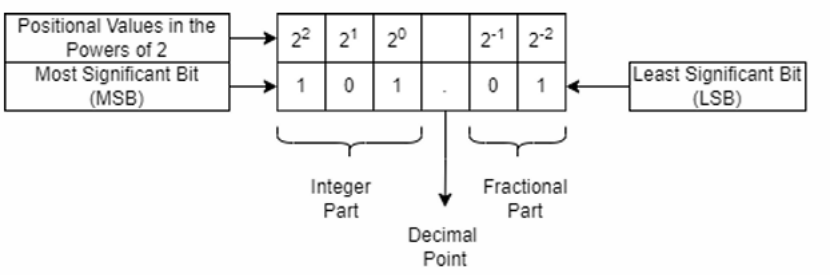
Fractional numbers can also be represented using similar notation, for example, 123.238 can be represented as

$$123.238 = 1 \times 10^2 + 2 \times 10^1 + 3 \times 10^0 + 2 \times 10^{-1} + 3 \times 10^{-2} + 8 \times 10^{-3}$$

The powers of 10, which correspond respectively to the digits in a decimal integer when read from right to left, are called the place values of the digits.

b. Binary System

The binary number system is base 2 and therefore requires only two digits, 0 and 1. The binary system is useful for computer programmers, because it can be used to represent the digital on/off method in which computer chips and memory work.



In the figure above, representation of the binary number (101.01)2 is shown. The most significant bit (MSB) is the bit in a binary number sequence with the largest value. This is usually the bit farthest to the left. The MSB represents the highest-order place of the binary integer. On the other hand, the Least Significant Bit (LSB) is the right-most bit of the number.

c. Hexadecimal System

The hexadecimal number system uses 16 unique symbols, 0-9 and A-F for representation of numbers. The place values in the hexadecimal system are powers of 16. Each hexadecimal digit has a unique 4-bit representation since $2^4 = 16$, so all possible digits can be represented with 4 bit binary numbers. The hexadecimal numbering system is often used by programmers to simplify the binary numbering system since large binary numbers can be organized and written neatly in their respective hexadecimal notations.

Hexadecimal Digit	Decimal Number	Binary Equivalent
0	**0**	**0000**
1	**1**	**0001**
2	**2**	**0010**
3	**3**	**0011**
4	**4**	**0100**
5	**5**	**0101**
6	**6**	**0110**
7	**7**	**0111**
8	**8**	**1000**
9	**9**	**1001**
A	**10**	**1010**
B	**11**	**1011**
C	**12**	**1100**
D	**13**	**1101**
E	**14**	**1110**
F	**15**	**1111**
10	**16**	**10000**
F0	**240**	**11110000**
FF	**255**	**11111111**

Some popular uses of Hexadecimal numbers are:

- Representation of addresses in memory.
- Representation of colors on web pages.
- Representation of Media Access Control (MAC) addresses.
- Display error messages.

d. Octal System

Octal system, or the base-8 system uses 8 unique symbols, 0, 1, 2, 3, 4, 5, 6, and 7. The place values in the octal system are powers of 8. We can use 3-bit binary numbers to represent the octal digits, since $2^3 = 8$.Converting from octal to binary proceeds in the same fashion as it does for hexadecimal:

Decimal Number	3-bit Binary Number	Octal Number
0	000	0
1	001	1
2	010	2
3	011	3
4	100	4
5	101	5
6	110	6
7	111	7

For example, an octal number $(532.67)_8$ can be represented in the decimal number system as

$$(532.67)_8 = 5 \times 8^2 + 3 \times 8^1 + 2 \times 8^0 + 6 \times 8^{-1} + 7 \times 8^{-2}$$

1.1 Binary Arithmetic

a. Addition

Rules for addition:

0+0=0; 0+1=1; 1+0=1; 1+1=10; 1+1+1=11

i. 100 + 01 = 101 …….(4 + 1 = 5)
ii. 1101 + 101 = 10010 ……. (13 + 5 = 18)
iii. 1111 + 110 = 10101 …….. (15 + 6 = 21)

b. Multiplication

Rules for multiplication:

0×0=0; 0×1=0; 1×0=0; 1×1=1

i. 101× 10 = 1010 …….. (5 × 2 = 10)
ii. 111 × 101 = 100011 ……. (7 × 5 = 35)
iii. 11011 × 1101 = 101011111 …… (27 × 13 = 351)

c. Subtraction

i. 10101 − 100 = 10001 (21 − 4 = 17)
ii. 1010 − 101 = 101 ….. we borrow a digit worth 2 to continue with the calculation (10 − 5 = 5)
iii. 1001011 − 110101 = 10110 (75 − 53 = 22)

1.2. Conversion From One Base to Another

a. Decimal System to Binary System

Steps for the Integer part

1. For the integral part of the number, repeatedly divide the number by 2, and save the remainder.

2. Repeat the process until no further division is possible.

3. Read the remainders from the bottom to top to get the binary number.

Steps for the fractional part

1. Repeatedly multiply the fractional part by 2 and note the integer and fractional part of the product.

2. Repeatedly multiply the fraction part by 2 and note the integer part from top to the bottom. (Stop if you get 0)

3. Repeat this procedure till sufficient precision is reached (usually 2 to 3 places after the radix point).

Example: Convert decimal number 75 to binary number.

Divisor	Dividend	Remainder	
2	75		
2	37	1	LSB
2	18	1	
2	9	0	
2	4	1	
2	2	0	
2	1	0	
	0	1	MSB

$=(1001011)_2$

Convert decimal number 142.67 to a binary number.

Divisor	Dividend	Remainder	
2	142		
2	71	0	LSB
2	35	1	
2	17	1	
2	8	1	
2	4	0	
2	2	0	
2	1	0	
	0	1	MSB

.67 x 2 = 1.34	1	.34
.34 x 2 = 0.68	0	.68
.68 x 2 = 1.36	1	.36
.36 x 2 = 0.72	0	.72

= $(10001110.1010)_2$

b. Binary System to Decimal System

Steps:

1. Multiply each digit with its respective weighted power of 2.

2. Calculate its sum.

Example: Convert Binary Number 101101 into Decimal Number.

$101101 = 1x2^5 + 0x2^4 + 1x2^3 + 1x2^2 + 0x2^1 + 1x2^0$

$= 32 + 0 + 8 + 4 + 0 + 1$

$= 45$

Convert Binary Number 110101.010 into Decimal Number.

$110101.010 = 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3}$

$= 32 + 16 + 0 + 4 + 0 + 1 + 0 + 0.25 + 0$

$= 53.25$

c. Binary System to Hexadecimal System

There are two ways to convert a binary number to a hexadecimal number:

1. Convert binary number to decimal number, and then convert to hexadecimal number.

2. Using grouping of bits.

For the integer part, make groups of four bits together starting from the left side of the radix point, and for the fractional part, make groups of four bits towards the right side of the radix point.
Replace each group with the corresponding hexadecimal number, we get the hexadecimal equivalent of the given binary number.
NOTE: Add any number of 0's in the left most bit for integer part, and 0's in right most bit for fractional part required to complete the grouping of 4 bits
Example: Convert binary number 1000110101110 to hexadecimal number.

*** Converting Binary Number into Decimal Number:**

$(1000110101110)_2 = 1 \times 2^{12} + 0 \times 2^{11} + 0 \times 2^{10} + 0 \times 2^9 + 1 \times 2^8 + 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$

$= 4096 + 0 + 0 + 0 + 256 + 128 + 0 + 32 + 0 + 8 + 4 + 2 + 0$

$= (4526)_{10}$

***Converting Decimal into Hexadecimal Number:**

Divisor	Dividend	Remainder	
16	4526		
16	282	14(E)	LSB
16	17	10(A)	
16	1	1	
	0	1	MSB

$= (11AE)_{16}$

Example 2: Convert Binary number 10010111101101 into Hexadecimal Number.

0010	0101	1110	1101
2	5	E	D

$= (25ED)_{16}$

Convert Binary number 101100100.100111 into Hexadecimal Number.

0001	0110	0100	.	1001	1100
1	**6**	**4**		**9**	**C**

Hence the equivalent Hexadecimal Number is $(164.9C)_{16}$

d. Hexadecimal System to Binary System

A hexadecimal number can be converted to Binary Number through any of the two methods:

1. Converting Hexadecimal Number to Decimal Number, and then converting Decimal Number to Binary number.

2. Using Conversion Table

Example: Convert Hexadecimal Number 7B316 to Binary Equivalent Number.

7	B	3	1	6
0111	1011	0011	0001	0110

= 0111 1011 0011 0001 0110

Hence the equivalent binary number is $(01111011001100010110)_2$

2. Data Representation in a Computer

Data Representation refers to the methods used internally to represent information stored in a computer. Computers store lots of different types of information:

- numbers
- text
- graphics of many varieties (stills, video, animation)
- sound

At least, these all seem different to us. However, all types of information stored in a computer are stored internally in the same simple format: a sequence of 0's and 1's. How can a sequence of 0's and 1's represent things as diverse as your photograph, your favorite song, a recent movie, and your term paper?

- Numbers must be expressed in binary form following some specific standard.
- Character data are assigned a sequence of binary digits

a. Unsigned Integers

Unsigned integers can represent zero and all positive integers. The value of an unsigned integer is interpreted as "the magnitude of its underlying binary pattern". That is, for an unsigned integer, all the bits in the binary pattern are used to represent the magnitude of the integer.

In binary, an n-bit pattern can represent 2^n distinct integers. Therefore, an n-bit pattern will represent integers from 0 to $2^n - 1$.

For example:

- A 4-bit pattern will represent the integers 0 to $(2^4) - 1 = 15$
- An 8-bit pattern will represent the integers 0 to $(2^8) - 1 = 255$

b. Signed Integers

- The digital computer handle both positive and negative integer.
- It means, is requiredfor representing the sign of the number (-or +),butcannot use the sign (-)to denote the negative number or sign (+) to denote the positive number.
- So, this is done by adding the leftmost bit to the number calledsign bit.
- A positive number has a sign bit 0, while the negative has a sign bit 1.
- It is also called asfixed point representation.
- A negative signed integer can be represented in one of the following:

1) Sign and magnitude method

2) One's complement method

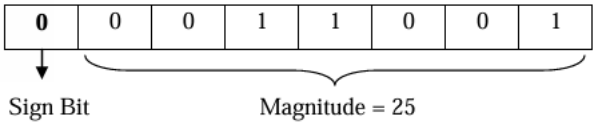
3) Two's complement method

➢ **Sign and magnitude method**

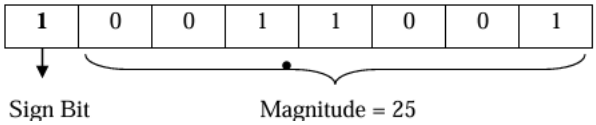
- An integer containing a sign bit followed by magnitude bits are calledsign-magnitude integer.
- In this method, first bit (MSB) is considered as a sign bit and the remaining bits stand for magnitude.
- Here positive number starts with 0and negativenumber starts with 1.

Example: Consider a decimal number $25_{(10)}$

- So the binary number is 11001(2). If we take the size of the word is 1 byte, then the number 25 will be represented as



- Suppose, if the number is-25, and then it will be represented as:



- **1's Complement representation**
  - This is the simplest method of representing negative binary number.
  - The 1's complement of a binary number is obtained by changing each 0 to 1 and each 1 to 0.
  - In other words, change each bit in the number to its complement.

Example1: Find the 1's complement of 101000

Original binary number	→	1 0 1 0 0 0
		↓ ↓ ↓ ↓ ↓ ↓
Find 1's Complement	→	0 1 0 1 1 1

Thus 1's complement of 101000 is 010111

Example2: Find the 1's complement of1010111.

- **2's Complement representation**
  - The 2's complement of a binary number is obtained bytaking 1's complement of the number and adding 1 to the Least Significant Bit (LSB) position.
  - The general procedure to find 2's complement is given by:

2's Complement = 1's Complement + 1

Example1: Find the 2's complement of 101000 and 1011101.

Note:

Negation: It is the operation of converting a positive number to its negative equivalent or a negative number to its positive equivalent. Negation is performed by performing 2's complement system.

Example: Consider the number +12. Its binary representation is 01100(2). Find the 2's complement of +12

Original binary number	0 1 1 0 0
Find 1's Complement	1 0 0 1 1
Add 1 to LSB + 1	+ 1
2's Complement	1 0 1 0 0

Clearly, this is a negative number since the sign bit is 1. Actually, 10100 represent-$12_{(10)}$ which is the negative equivalent of the number $12_{(10)}$

➢ **Subtraction of Binary Number using Complement**

- Most of the computers perform subtraction using complemented number. This is less expensive because the same addition circuit is used for subtraction with slight changes in the circuit.
- In the binary number system we can perform subtraction operation using two methods of complements:
  - Subtraction using 1's Complement

***Case 1**: Subtracting a smaller number from a larger number (Minuend is greater than Subtrahend)

Step 1: Find the 1's complement of the subtrahend.
Step 2: Add this to the minuend.
Step 3: Carry is generated, this carry is called as the end around carry
Step 4: Add the end around carry back to the LSB to get the final difference

Example1: Subtract 15 from 23 using 1's complement.

	Decimal	Binary
Minuend	23	10111
Subtrahend	-15	01111

1's complement of subtrahend is 10000

Minuend	→	1 0 1 1 1
1's Complement of subtrahend	→	+ 1 0 0 0 0
End around carry	→	1 1 0 1 1 1
Add end around carry	→	+ 1
Difference is	→	1 0 0 0

***Case2**: Subtracting a larger number from a smaller number (Minuend is less than Subtrahend)

Step 1: Find the 1's complement of the subtrahend.
Step 2: Add this to the minuend.
Step 3: There will be no carry, Re complement the answer to get the difference

Example1: Subtract 52 from 25 using 1's complement.

	Decimal	Binary
Minuend	25	011001
Subtrahend	-52	110100

1's complement of subtrahend is 001011

Minuend	→	0 1 1 0 0 1
1's Complement of subtrahend	→	+ 0 0 1 0 1 1
	→	1 0 0 1 0 0
Since there is no carry		↓ ↓ ↓ ↓ ↓ ↓
take 1's complement and	→	0 1 1 0 1 1
attach a negative sign		Hence, the result = - 011011 i.e. - 27

- Subtraction using 2's Complement

***Case 1:** Subtracting a smaller number from a larger number (Minuend is greater than Subtrahend)

Step 1: Find the 2's complement of the subtrahend.

Step 2: Add this to the minuend.

Step 3: Carry is generated, Discard the carry and the remaining bits give the difference

- **Example 1**: Subtract 09 from 17 using 2's complement.

		Decimal	Binary	
Minuend		17	10001	
Subtrahend		-09	01001	
1's complement of subtrahend (9) is			1 0 1 1 0	
Add 1 to LSB			+ 1	**STEP 1**
2's complement of 9 is			1 0 1 1 1	
Minuend	→		1 0 0 0 1	
2's Complement of subtrahend	→		+ 1 0 1 1 1	**STEP 2**
End around carry	→		1 0 1 0 0 0	
			Discard the carry	**STEP 3**
Difference is	→		1 0 0 0	

Hence, the result = 1000 i.e. 8

***Case2:** Subtracting a larger number from a smaller number (Minuend is less than Subtrahend)

Step 1: Find the 2's complement of the subtrahend.

Step 2: Add this to the minuend.

Step 3: There will be no carry, hence take the 2's complement of the answer and place a negative sign in front.

- **Example 1**: Subtract 47 from 26 using 2's complement.

		Decimal	Binary	
	Minuend	26	011010	
	Subtrahend	-47	101111	
	1's complement of subtrahend (47) is		0 1 0 0 0 0	**STEP 1**
	Add 1 to LSB		+ 1	
	2's complement of 9 is		0 1 0 0 0 1	
	Minuend	→	0 1 1 0 1 0	**STEP 2**
2's Complement of subtrahend		→	+ 0 1 0 0 0 1	
			1 0 1 0 1 1	
1's Complement of answer		→	0 1 0 1 0 0	**STEP 3**
2's Complement of answer		→	0 1 0 1 0 0	
			+ 1	
			0 1 0 1 0 1	

Hence, the result = 10101 i.e. 21

VI. Computer Codes

Computer code helps us to represent characters in a coded form in the memory of the computer. These codes represent specific formats which are used to record data. Some of the commonly used computer codes are:

- Binary Coded Decimal (BCD)
- Extended Binary Coded Decimal Interchange Code (EBCDIC)
- American Standard Code for Information Interchange (ASCII)
- Excess-3 Code

1. BCD code (or Weighted BCD Code or 8421 Code)

BCD stands for Binary Coded Decimal. It is one of the early computer codes.In this coding system, the bits are given from left to right, the weights 8,4,2,1 respectively.

The BCD equivalent of each decimal digit is shown in table.

Decimal Digit	BCD Equivalent	Decimal Digit	BCD Equivalent
0	0000	5	0101
1	0001	6	0110
2	0010	7	0111
3	0011	8	1000
4	0100	9	1001

- **Example**: Convert the decimal number 537 into BCD.

5 3 7

0101 0110 0111 Hence, the BCD of 537 is 010101100111

In 4-bit BCD only 2^4=16 configurations are possible which is insufficient to represent the various characters

2. Excess-3 BCDcode (or XS-3 Code)

- The Excess-3 BCD code is a non-weighted code used to express decimal number.
- The name Excess-3 code derived from the fact that each binary code is the corresponding BCD code plus $0011_{(2)}$(i.e. Decimal 3).
- This code is used in some old computers.
- The following table gives the Excess-3 code equivalent of decimal (0-9)

Decimal Number	BCD Number	Adding 3 to BCD	Excess-3 Equivalent
0	0000	11	0011
1	0001	11	0100
2	0010	11	0101
3	0011	11	0110
4	0100	11	0111
5	0101	11	1000
6	0110	11	1001
7	0111	11	1010
8	1000	11	1011
9	1001	11	1100

- **Example**: Find Excess-3(XS3) representation of decimal number 537.

5	3	7	→	Decimal Digit
0101	0110	0111	→	8421 BCD Code
0011	0011	0011	→	Add 3
1000	1010	0110	→	Excess-3 Code

3. EBCDIC

- It stand for Extended Binary Coded Decimal Interchange Code.
- This was developed by IBM.
- It uses an 8-bit code and hence possible to represent 256 different characters or bit combinations
- EBCDIC is used on most computers and computer equipment today.
- It is a coding method generally used by larger computers(mainframes)to present letters, numbers or other symbols in a binary language the computer can understand.
- EBCDIC is an 8-bit code; therefore, it is divided into two 4-bit groups, where each 4-bit can be represented as 1 hexadecimal digit.

4. ASCII

- It standsfor theAmerican Standard Code for Information Interchange.
- It is a 7-bit code, which is possible to represent 2^7=128 characters.
- It is used in most microcomputers and minicomputers and in mainframes.
- The ASCII code is of two types–ASCII-7 and ASCII-8.
- ASCII-7 is 7-bit code for representing English characters as numbers, with each letter assigned a number from 0 to 127.

Example: The ASCII code for uppercase M is 77.

Dec	Bin	Hex	Char	Dec	Bin	Hex	Char	Dec	Bin	Hex	Char	Dec	Bin	Hex	Char
0	0000 0000	00	[NUL]	32	0010 0000	20	space	64	0100 0000	40	@	96	0110 0000	60	`
1	0000 0001	01	[SOH]	33	0010 0001	21	!	65	0100 0001	41	A	97	0110 0001	61	a
2	0000 0010	02	[STX]	34	0010 0010	22	"	66	0100 0010	42	B	98	0110 0010	62	b
3	0000 0011	03	[ETX]	35	0010 0011	23	#	67	0100 0011	43	C	99	0110 0011	63	c
4	0000 0100	04	[EOT]	36	0010 0100	24	$	68	0100 0100	44	D	100	0110 0100	64	d
5	0000 0101	05	[ENQ]	37	0010 0101	25	%	69	0100 0101	45	E	101	0110 0101	65	e
6	0000 0110	06	[ACK]	38	0010 0110	26	&	70	0100 0110	46	F	102	0110 0110	66	f
7	0000 0111	07	[BEL]	39	0010 0111	27	'	71	0100 0111	47	G	103	0110 0111	67	g
8	0000 1000	08	[BS]	40	0010 1000	28	(	72	0100 1000	48	H	104	0110 1000	68	h
9	0000 1001	09	[TAB]	41	0010 1001	29	)	73	0100 1001	49	I	105	0110 1001	69	i
10	0000 1010	0A	[LF]	42	0010 1010	2A	*	74	0100 1010	4A	J	106	0110 1010	6A	j
11	0000 1011	0B	[VT]	43	0010 1011	2B	+	75	0100 1011	4B	K	107	0110 1011	6B	k
12	0000 1100	0C	[FF]	44	0010 1100	2C	,	76	0100 1100	4C	L	108	0110 1100	6C	l
13	0000 1101	0D	[CR]	45	0010 1101	2D	-	77	0100 1101	4D	M	109	0110 1101	6D	m
14	0000 1110	0E	[SO]	46	0010 1110	2E	.	78	0100 1110	4E	N	110	0110 1110	6E	n
15	0000 1111	0F	[SI]	47	0010 1111	2F	/	79	0100 1111	4F	O	111	0110 1111	6F	o
16	0001 0000	10	[DLE]	48	0011 0000	30	0	80	0101 0000	50	P	112	0111 0000	70	p
17	0001 0001	11	[DC1]	49	0011 0001	31	1	81	0101 0001	51	Q	113	0111 0001	71	q
18	0001 0010	12	[DC2]	50	0011 0010	32	2	82	0101 0010	52	R	114	0111 0010	72	r
19	0001 0011	13	[DC3]	51	0011 0011	33	3	83	0101 0011	53	S	115	0111 0011	73	s
20	0001 0100	14	[DC4]	52	0011 0100	34	4	84	0101 0100	54	T	116	0111 0100	74	t
21	0001 0101	15	[NAK]	53	0011 0101	35	5	85	0101 0101	55	U	117	0111 0101	75	u
22	0001 0110	16	[SYN]	54	0011 0110	36	6	86	0101 0110	56	V	118	0111 0110	76	v
23	0001 0111	17	[ETB]	55	0011 0111	37	7	87	0101 0111	57	W	119	0111 0111	77	w
24	0001 1000	18	[CAN]	56	0011 1000	38	8	88	0101 1000	58	X	120	0111 1000	78	x
25	0001 1001	19	[EM]	57	0011 1001	39	9	89	0101 1001	59	Y	121	0111 1001	79	y
26	0001 1010	1A	[SUB]	58	0011 1010	3A	:	90	0101 1010	5A	Z	122	0111 1010	7A	z
27	0001 1011	1B	[ESC]	59	0011 1011	3B	;	91	0101 1011	5B	[	123	0111 1011	7B	{
28	0001 1100	1C	[FS]	60	0011 1100	3C	<	92	0101 1100	5C	\	124	0111 1100	7C	|
29	0001 1101	1D	[GS]	61	0011 1101	3D	=	93	0101 1101	5D	]	125	0111 1101	7D	}
30	0001 1110	1E	[RS]	62	0011 1110	3E	>	94	0101 1110	5E	^	126	0111 1110	7E	~
31	0001 1111	1F	[US]	63	0011 1111	3F	?	95	0101 1111	5F	_	127	0111 1111	7F	[DEL]